Warmer and longer growing seasons increase growth for juvenile trees in a common garden

Christophe Rouleau-Desrochers[1], Victor Van der Meersch[1], D.M. Buonaiuto[2,3,5], C.J. Chamberlain[2,3,4], Deirdre Loughnan[1], Neil Pederson, & E.M. Wolkovich[1,2,3]

[1]Forest & Conservation Sciences, Faculty of Forestry & Environmental Stewardship, University of British Columbia, Vancouver, British Columbia, Canada
[2]Arnold Arboretum of Harvard University, Boston, Massachusetts, USA
[3]Department of Organismic and Evolutionary Biology, Harvard University, Cambridge, Massachusetts, USA
[4]The Nature Conservancy, USA
[5]Department of Plant Science and Landscape Architecture, University of Maryland, College Park, Maryland, USA

Abstract

Anthropogenic climate change affects natural systems at the global scale, with the most frequently observed biological impact of climate change being shifts in phenology—the timing of recurring life history events. These shifts will cause other cascading effects that are likely to further feedback to future consequences on the climate. For trees, shifted phenology has extended the vegetative growing season, which is widely expected to increase tree growth, with impacts on forest carbon sequestration. While decades of work has assumed longer seasons increase tree growth, multiple recent studies have found no connections. Here, we used juvenile tree data to test the relationships between growing season length and growth in a common garden where both drought and competition were low. Across species within the *Betulaceae* family sourced from four provenances we found increased tree growth with warmer seasons (three out of four species), and with longer seasons, but for fewer species (two out of four species). Our results suggest growing conditions may be more important than season length alone. We found little effect of provenance on growth or phenology, including end of season events—though end of season phenology was more connected to growth responses than start of season. Taken together, our results suggest thermal conditions across the season may predict growth responses and vary considerably across species, but not provenance. Our work has implications for forest regeneration, especially given our focus on juvenile trees, and could help improve vegetation models.

Introduction

At the global scale, shifts in phenology—the timing of recurring life history events–are a direct consequence of anthropogenic climate change (Parmesan *et al.*, 1999; Menzel *et al.*, 2006; Calvin *et al.*, 2023). Across many plant species, spring events have advanced (Wolfe *et al.*, 2005; Chmielewski & Rötzer, 2001; Fu *et al.*, 2014), while autumn events have delayed—but to a lesser extent (Gallinat *et al.*, 2015; Jeong & Medvigy, 2014). Together, early springs and later autumns lengthen the growing season, which should increase growth for many species, including trees (Keenan *et al.*, 2014; Stridbeck *et al.*, 2022).

How much trees increase growth with longer seasons is important given the major role of forests in carbon sequestration (Friedlingstein *et al.*, 2026). Temperate forests account for roughly 40% of the annual global carbon sink, diminishing the consequences of greenhouse gas emissions on the climate (Gibbs *et al.*, 2025). With warming, research has repeatedly connected longer seasons and increased carbon storage (Boisvenue & Running, 2006; Gao *et al.*, 2022; Finzi *et al.*, 2020).

The future of the temperate forest carbon sink relies on the assumption that longer and warmer growing seasons will continue to support carbon uptake and storage by trees (Pan *et al.*, 2024). If this relationship weakens under ongoing climate change, the capacity of forests to sequester carbon could decline, affecting projections of the terrestrial carbon sink. Such changes are particularly relevant for forest restoration efforts that depend on sustained tree growth and survival to deliver long-term climate mitigation benefits (Cook-Patton *et al.*, 2020). Because successful restoration increasingly requires selecting species and species mixtures that can thrive under warmer temperatures and greater drought stress, improved models of forest growth, mortality, and carbon dynamics are needed to guide restoration planning and optimize carbon storage across land ecosystems (Ruehr *et al.*, 2023; Chang *et al.*, 2024).

New studies suggest that longer growing seasons do not always increase growth (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Matula *et al.*, 2023; Charlet De Sauvage *et al.*, 2022; Cabon *et al.*, 2022). Recently, Dow *et al.* (2022) found that warmer springs lengthened the growing season, but did not increase the peak growth rate or total annual growth in two temperate forest communities. Working across biomes, Cabon *et al.* (2022) found a similar trend, where gross primary production and growth appeared decoupled. However, neither of these studies could clearly establish why they found no relationship while previous studies have. Understanding whether temperate forests will persist as a net carbon sink requires answering why trees do not grow more despite longer growing seasons.

A longer growing season may not mean more favorable thermal conditions for growth. The well-documented advance of spring phenology elongates the growing season (Keenan *et al.*, 2014), but when spring events occur earlier, days are short and usually cool (Figure 1a). Therefore, this advance may contribute little to a tree's annual growing degree days (GDD) accumulation (Figure 1b). In contrast, even small delays in the timing of end of season events can lead to larger changes in annual GDD; since more GDD accumulates per day during this period, any extra days at the end of season may be more important than additional days at the start of the season (Buonaiuto *et al.*, 2026).

Local adaptation could limit tree growth responses of different populations to future climate change (Soolanayakanahally *et al.*, 2013; Wolkovich *et al.*, 2025). Tree growth may be internally constrained, as it—like other biological processes—is under strict genetic control. Even with optimal environmental conditions, physiological constraints may prevent temperate trees from growing indefinitely (Körner *et al.*, 2023; Wolkovich *et al.*, 2025), but we currently do not understand well how tree phenology and growth can track changing conditions. While leaf phenology (e.g. leafout) in the spring is plastic and driven by temperature, end of season phenology (e.g. budset) is less plastic and mostly driven by photoperiod (Baumgarten *et al.*, 2026; Singh *et al.*, 2017), with evidence of local adaptation (Zeng & Wolkovich, 2024). Provenance trials have repeatedly found trees from northern latitudes set bud earlier, even when grown in warm climates with longer growing seasons (Aitken & Bemmels, 2016; Way & Montgomery, 2015). This local adaptation and related reliance on photoperiod to trigger budset (an end of season event) likely explains why shifts in the end of season with warming have been so much smaller than shifts in the start of season (Piao *et al.*, 2019). This predicts a latitudinal gradient in growth responses, with more limited responses in poleward populations where the end of season is constrained to happen earlier in the season. Thus, common gardens investigating the effect of provenance latitude could help predict how climate change will affect tree growth.

To understand the relationship between growing seasons and growth, we need to understand how growth varies across species and populations while disentangling how longer seasons are also more thermally favorable seasons. Here, we address this by studying how growth and growing seasons relate across species, populations and multiple metrics of growing season length. Working with juvenile trees sourced from four populations and grown in one common garden, we examine how growth, measured via tree rings, and leaf phenology covary over three years. We consider effects of both the length of the season (i.e. number of growing days in the season) and its thermal conditions (i.e. GDD accumulated in the season).

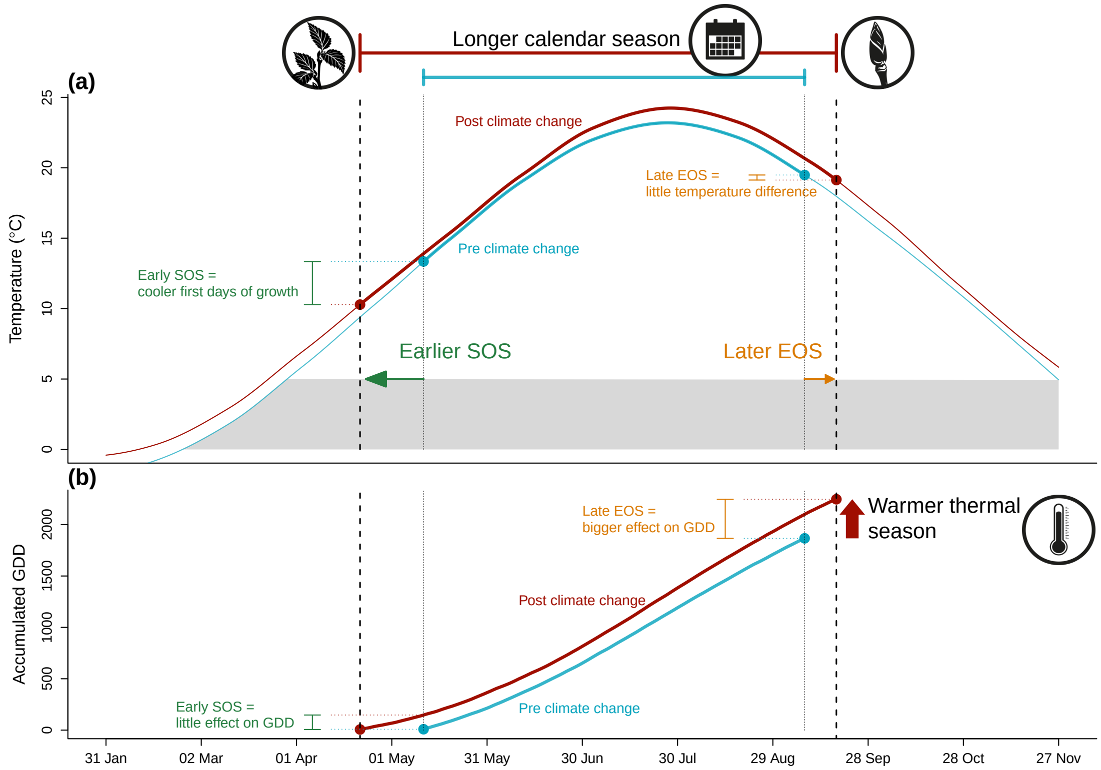

Figure 1: Shifting growing season lengths with climate change affect both the temperature trees experience throughout the growing season (a), and how many growing degree days (GDD) they accumulate (b). Here we show changes in daily (smoothed) and accumulated temperatures before (1945-1965, in blue) and significant anthropogenic climate warming (2005-2025, in red) using data local to our study. The arrows represent hypothetical shifts in spring (green) and autumn (orange) phenology. With climate change, spring events have advanced and autumn phenology has delayed, but to a much lesser extent—lengthening the calendar growing season. This concurrently changed the temperature trees experience early in their season, with cooler first days of growth (a) causing little accumulation of GDD (b). In contrast, weaker shifts in the end of season and little temperature change (a) result in a relatively high GDD accumulation (b) due to warmer temperature at this time of the season. Alongside these start and end of season shifts, warmer summer temperatures increased GDD accumulation, further warming their thermal seasons. We used mean daily temperature data from the Boston Logan International Airport Climate Station for the smoothed temperature and GDD curves (National Oceanic and Atmospheric Administration (NOAA), 2026).

Materials and Methods

Common garden setup and data collection

We collected seeds from 18 species of deciduous trees and shrubs from four provenances across a 3.5° latitudinal gradient (Figure 2; Table ??) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings (Baskin & Baskin, 2014). In the spring of 2017, we planted the seedlings outside at the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned in the autumn of 2020 (Buonaiuto *et al.*, 2026).

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December; the exact frequency varied within and across years (for more details, see Figure S2). For this, we used a modified BBCH scale to observe two phenological stages: leafout which we defined as BBCH 15 and budset as BBCH 102; (Finn *et al.*, 2007; Buonaiuto *et al.*, 2026). We had a total of 239 leafout and 230 budset observations (which we refer to as the 'full datasets'), but for some individual trees in certain years we did not have both events; thus our total of matched full growing season duration (which we refer to as the 'restricted dataset') is 227 observations.

In the spring of 2023, we collected tree ring samples from tree species with high survivorship. Thus, we have four species in the *Betulaceae* family : grey alder (*Alnus incana ssp. rugosa*), yellow birch (*Betula alleghaniensis*), paper birch (*Betula papyrifera*) and grey birch (*Betula populifolia*). We collected cross-sections on 78 individuals by cutting the trunk above the root collar, and cores on 43 individuals using standard dendrochronological methods (Cook & Kairiukstis, 1990), for a total of 81 individual trees (thus, we had three individuals with cores only; Table ??). We stored the cores in modified plastic straws to promote aeration and left them to dry with the cross-sections at ambient temperature for three months.

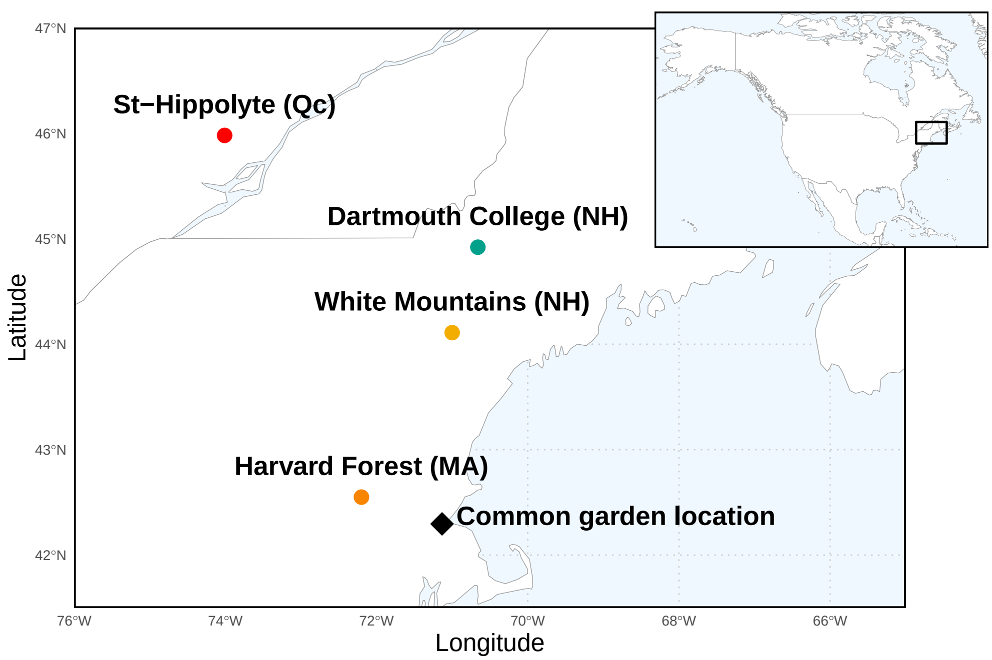

Figure 2: Provenance where the seeds were sourced from is depicted as circles, while the diamond denotes the location of the common garden. The common garden location is where the study took place at the Weld Research Building (Boston, MA). The colors of the dots match the model estimates of Figure S1.

Sample processing, imaging and measuring

We mounted cores on wooden mounts and sanded both the cores and cross-sections using progressively finer sandpaper grits (150, 300, 400, 600, 800, 1000), then scanned the cores and cross-sections at a resolution of 6250 dpi with a high-resolution tree ring scanner (Fong *et al.*, 2026). Using the digitized images, we measured ring widths with ImageJ (Schneider *et al.*, 2012) at three standard locations for each cross-section (at 120° from each other, averaging the three measurements to yield one ring width per year, per individual), and one ring width for each core. When we had cross-sections ($n = 78$) and cores ($n = 43$) for an individual, we used only the measurement from the cross-sections ($n = 81$).

We performed visual cross dating. This consists of assigning a calendar year to each ring and comparing the year-to-year growth variation and anomalies across replicates of the same species (Cook & Kairiukstis, 1990). We did not perform statistical crossdating or detrending because our chronologies (number of years of data) were extremely short (Raden *et al.*, 2020) and showed no evidence of consistent allometric trends (Figure S3). We controlled for effects of different years by including 'year' in our models (see equation 1 in 'Data analyses' below).

Data analyses

To understand how growth shifted with season length, we used Bayesian hierarchical models, estimating how growth, measured as ring width, covaried with season length. We measured the season length in four different ways: GDD, GSL, SOS and EOS. Leaf phenology and growth are often closely synchronized in diffuse-porous species (e.g., *Betula;* Čufar *et al.*, 2008) making leaf phenology a good proxy for setting the start and end of the growing season in these species (Marchand *et al.*, 2021; Silvestro *et al.*, 2025; Suzuki *et al.*, 1996). Thus, we calculated the thermal season using growing degree days (GDD; mean daily temperature above 5°C) accumulated between leafout and budset, and the calendar season (or growing season length; GSL) as the number of days between leafout and budset. Finally, we used leafout for the start of season (SOS) and budset for the end of the season (EOS) as separate predictors that represent the boundaries of the growing season. For interpretability, we report ring width change in % with the number of GDDs accumulated in an average week (seven days) in May (day of year 121 to 150 from 2018 to 2020), for the thermal season, and ring width change in % per week of season length, leafout and budset for GSL, SOS and EOS, respectively. We report effects when the 90% UI do not overlap with zero and consider a trend if the 50% UI do not overlap with zero (**?**).

For daily climate data across all the years of phenological sampling (2018-2020), we combined two sources. For 2018 and 2019, we used data from the climate station located at Weld Hill research station (Arnold Arboretum, 2026). Because the data were not available for that station after August 2020, we used data from the 'Boston (Weld Hill)' weather station of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (see Figure S11 for the climate overlap between the two stations in 2020; Carroll *et al.*, 2026). We used the North American Drought Monitor Maps to determine drought occurrence across years at our study location and used the proposed drought intensity index to qualify the presence of moderate, severe or extreme droughts (National Centers for Environmental Information (NCEI), 2026).

For each predictor, we estimated its effect on observed ring width (rw_i, denoting each observation) for each tree (*treeid*) in each year (*year*) as:

$$\begin{aligned}
\log(\mathrm{rw}_i) &\sim Normal(\mu_i, \sigma_y) \\
\mu_i &= \alpha + \alpha_{\text{species}} + \alpha_{\text{prov}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor[i]} \\
\alpha_{\text{prov}} &\sim Normal(0, \sigma_{\alpha\,\text{prov}}) \\
\alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha\,\text{treeid}})
\end{aligned} \tag{1}$$

This model estimates a species-specific effect of each season length predictor (e.g. GDD) on growth ($\beta_{species}$), while also controlling for other variation due to species ($\alpha_{species}$), provenance (α_{prov}), individual tree (α_{treeid}) and year (α_{year}). We ran the models using both natural units, and also using scaled predictors (via *z*-

scoring. Gelman & Hill, 2006) to allow for direct comparison of effect sizes across different metrics. We report estimates as means and 90% uncertainty intervals (UI).

We used weakly informative priors for all models (increasing the priors by 2× for the z-scored models did not qualitatively affect the results). We ran four chains with 2000 warm-up and 2000 sampling iterations each, for a total of 8000 sampling iterations and checked output diagnostics: chains had no divergent transitions, $\hat{R}$ values below 1.01, and effective sample sizes (n_{eff}) greater than 10% of the model iterations. We also fit the models for SOS and EOS using both the restricted ($n = 227$) and full datasets for each predictor (SOS $n = 239$ and EOS $n = 230$). We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R (R Core Team, 2021).

In addition, we compared the relative effect of an extension in the SOS versus EOS using our estimated effect of GDD (Equation 1). To do this, we averaged the GDD (mean daily GDD from 2005 to 2025) accumulated seven days before the median leafout (for SOS) and after the median budset (for EOS), for each species, across all years of our dataset. Then, with our estimated parameters, we report the % effect of GDD on growth for an extended SOS and EOS: %Δring width = $(e^{\beta * GDD_{extSOS/EOS}} - 1) \times 100$.

Because ring widths decrease with age—particularly with young trees (CITES)—we also ran the models using basal area increment (BAI) as the response variable. This allows for a more representative portrayal of annual growth increment by using the total ring area, instead of its width alone. To calculate BAI we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pith to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring: $\pi \times (\text{inner radius}^2 - \text{outer radius}^2)$, and averaged across the three BAI for each ring.

Results

Relationship between growth and thermal and calendar growing season

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in % per one week unit (see 'Data analyses' section in Methods for more details). Warmer thermal seasons increased growth for paper birch (*Betula papyrifera*; β = 16.08% (UI: 6.26, 26.42)), grey birch (*Betula populifolia*; β = 7.23% (UI: 2.50, 12.22)), and yellow birch (*Betula alleghaniensis*; β = 4.27% (UI: 0.56, 8.08)). Grey alder (*Alnus incana*) trended in the same direction (β = 2.20% (UI: -1.08, 5.55)); Figure 3a and e; TableS3). Unlike its response to warmer thermal season, *A. incana* grew more with longer calendar seasons (β = 5.22% (UI: 0.18, 10.39)). *B. papyrifera* also grew more with longer calendar seasons (β = 20.67% (UI: 5.59, 36.64)), while the other two *Betula* did not, though they trended in the same way (*B. alleghaniensis*: β = 3.35% (UI: -2.95, 9.73), and *B. populifolia*: β = 5.74% (UI: -1.96, 13.87); Figure 3b and f; Table S4). On a scaled axis (through z-scoring, Gelman & Hill, 2006), the thermal and calendar season were similarly predictive (Figure S10). Results with basal area increment (BAI) for the GDD model were similar to ring width (see Table S7).

Growth at the start versus the end of the growing season

Increased growth with longer seasons is often presumed to come from growth in the spring, since spring phenology has shifted much more than autumn phenology. Yet, only one species grew more with an earlier start of season (*A. incana* (one of the two species for which a longer calendar season increased growth: β = -7.00% (UI: -12.51, -1.16); more growth with an earlier start of season is represented as a negative slope; Figure 3c and g; Table S4). *B. populifolia* trended in the opposite direction (β = 5.17% (UI: -6.55, 17.51); Figure 3c and g; Table S6), and *B. alleghaniensis*, which did not grow more with a longer calendar season, grew less with an earlier start of season (β = 10.63% (UI: 0.34, 21.50); Figure 3c and g; Table S5). In contrast, we found most species grew more with a later end of season (*B. alleghaniensis*: β = 12.22% (UI: 5.16, 19.41), *B. populifolia*: β = 10.03% (UI: 2.14, 18.09), and *B. papyrifera*: β = 23.74% (UI: 9.37, 39.20); Figure 3d and h; Table S6), with the latter being the only one of those three that also grew more with longer calendar season. The estimates were unchanged when using the full (SOS: $n = 239$ and EOS: $n = 230$) or restricted datasets ($n = 227$; Figure S4a).

More growth with a later EOS could be due to the warmer days at the end of the season. Supporting this, we found that the addition of an estimated 7 days at the EOS increased growth across all species by 10.69% (UI: 6.11, 15.48), compared to an increase of 7.16% (UI: 4.10, 10.31) from an earlier SOS (estimated using our model of GDD shown in Figure 3, see 'Data analyses' section in Methods for more details; Table S8). Differences in magnitude between an extension of SOS versus EOS were similar across species (Table S8).

Year and provenance level effects on growth

We did not find any clear effect of year on growth (2018: $\alpha = 0.17$ (UI: -0.77, 1.14) ring width (log(mm); henceforth 'RW'); 2019: $\alpha = 0.16$ (UI: -0.78, 1.12) RW; 2020: $\alpha = -0.45$ (UI: -1.39, 0.51) RW; Table S3), and there were no apparent differences in baseline growth between species (*A. incana*: $\alpha = 1.01\%$ (UI: -3.14, 5.18) RW; *B. alleghaniensis*: $\alpha = 0.15\%$ (UI: -4.04, 4.24) RW; *B. papyrifera*: $\alpha = -4.04\%$ (UI: -9.02, 0.89) RW; *B. populifolia*: $\alpha = -0.70\%$ (UI: -5.04, 3.50) RW; Figure S5 and S6; Table S3). In addition, growth did not vary with provenance (Dartmouth College (NH): $\alpha =$ (UI: ,) RW; Harvard Forest (MA): $\alpha =$ (UI: ,) RW; St-Hippolyte (Qc): $\alpha =$ (UI: ,) RW; White Mountains (NH): $\alpha =$ (UI: ,) RW; Figure S1; Table S3). The limited variation between provenances ($\sigma = 0.18$ (UI: 0.02, 0.48 RW) was similar to variation across individual trees ($\sigma = 0.17$ (UI: 0.05, 0.27) RW; Table S3).

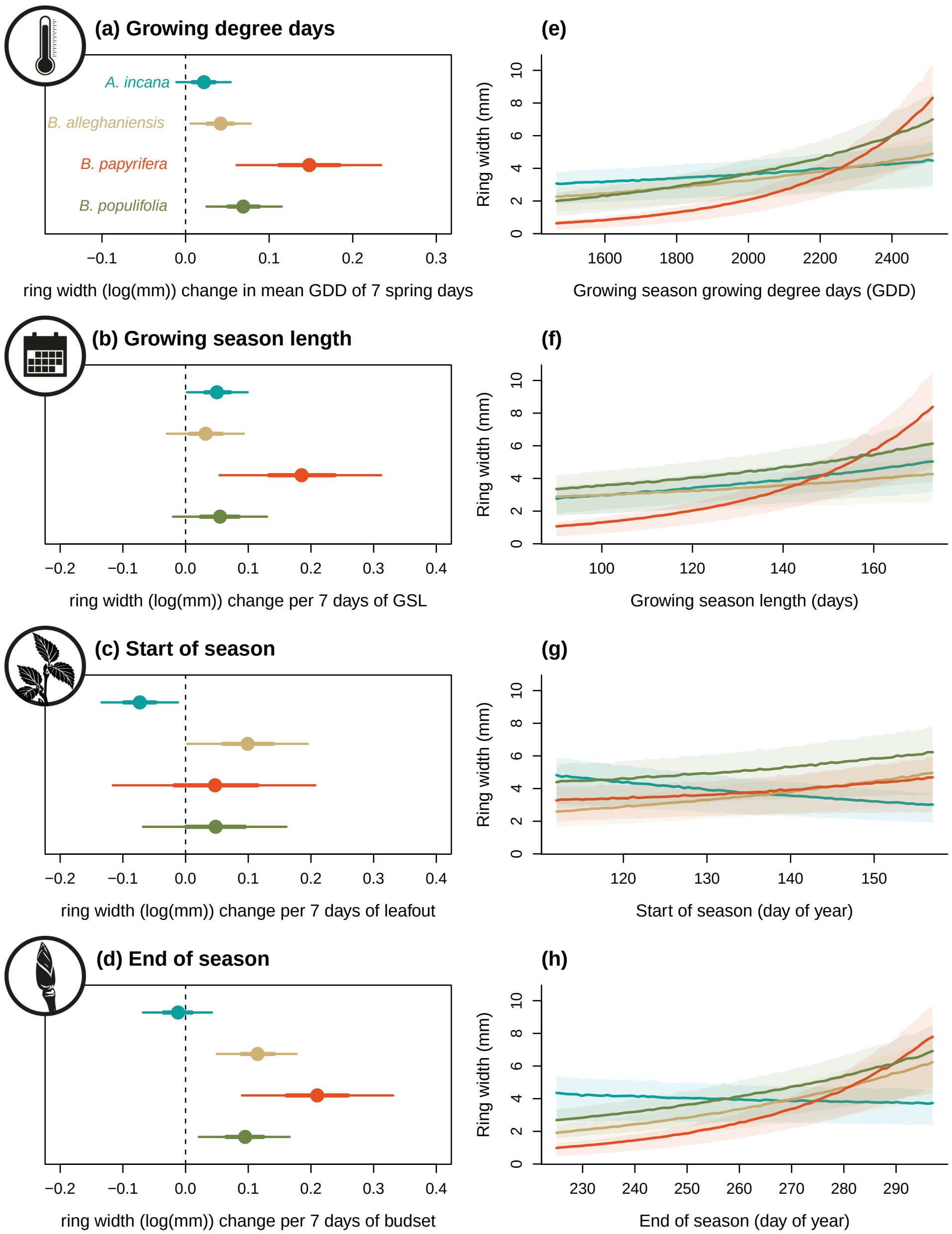

Figure 3: Estimates of ring width change as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length, start of season, and end of season across four species: grey alder (*Alnus incana*), yellow birch (*Betula alleghaniensis*), paper birch (*Betula papyrifera*) and grey birch (*Betula populifolia*; species colors are the same across the panels). Panels on the left (a to d) represent the mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor, which we represent through ring width change in mm on the log scale per adjusted predictor (see 'Data analyses' section in Methods for more details on predictor adjustment). Panels on the right (e to h) show the estimated response of ring width (mm) as a function of each predictor, with the lines depicting the mean response at each predictor value, and the shaded area, the 50% uncertainty intervals. See Tables S3, S4, S5 and S6 for parameter estimates.

Discussion

Here we used 81 juvenile trees of four species from four provenances to study how longer growing seasons affect tree growth (using annual growth via tree rings and leaf phenology as a proxy for the start and end of season). Longer growing seasons increased growth for all species, but varied depending on the metric used to measure season length, with the thermal growing season more often relating to growth than the calendar season.

Our results support the hypothesis that the thermal season is a better predictor for growth (Chuine & Régnière, 2017; Wang & Engel, 1998). By accounting for differences in daily temperatures, the thermal season may be more closely aligned to when and how fast plants can grow each day compared to the calendar season, which assumes a stable influence of each day on growth (Körner *et al.*, 2023). Thus, the short and cool spring days early in the season count for less in the thermal season than the calendar season (Buonaiuto *et al.*, 2026). Further supporting this, when we disentangled the calendar season by using the start and end of season separately, we found a larger response of growth to the end of season, with little effect of the start of season. The finding that the end of season appears more related to growth than the start of season is aligned with increasing work suggesting that the start and end of season together may dynamically determine growth (Zohner *et al.*, 2023). This implies that research focusing solely on the start of season (Dow *et al.*, 2022; Gao *et al.*, 2022; Wheeler *et al.*, 2016) could miss when longer seasons lead to more growth. Phenology research that strongly focuses on the start of season also implies that we may ignore an important part of the season for growth. While studies have long called for more research on the end of season (Gallinat *et al.*, 2015; Gunderson *et al.*, 2012; Piao *et al.*, 2019; Richardson *et al.*, 2012), our results suggest we may especially need more work that monitors how growth and phenology shift across the season (Zani *et al.*, 2020; Zohner *et al.*, 2023). More studies using data with a finer temporal scale (e.g., daily or weekly data through dendrometers and/or xylogenesis) could confirm if most growth occurs closer to the end versus the start of season. To date, these studies showed strong species differences in when most growth happens during the season. However, most of these studies were done on a small number of European and/or conifer species (e.g. Zweifel *et al.*, 2016; Etzold *et al.*, 2022; Balducci *et al.*, 2019), highlighting that we need more data across species spanning a wider range of life history strategies.

Our results suggest different growth strategies between species may yield different responses to longer seasons. Our most ruderal species, *A. incana* was the only species to grow more with a longer calendar season and an earlier start of season. This suggests that species with more ruderal, flexible life history strategies may actually take advantage of growing more in the spring despite cooler, shorter days (Bowman & Hacker, 2023; Furlow, 1997; Grime, 1977). In contrast, growth of *B. papyrifera* and *B. populifolia*, which are fast-growing but more conservative than *A. incana*, were both more driven by the end of season, suggesting that they may be able to exploit these late-season days characterized by warmer temperatures (Burns *et al.*, 1990). Lastly, our slowest growing, longest-living species, *B. alleghaniensis*, responded to both the start and end of season. This points towards a possible 'neutralizing' response where an early start of season decreases growth, but a later end of season increases it, offsetting this species' response to a longer calendar season.

Studies including an even wider span of growth strategies across species may find further differences. Our four study species all grow relatively fast, which should allow them to track changing conditions better than their slower-growing counterparts. While we found an overall positive trend of warmer seasons on growth, this might have changed if we used slower-growing, more conservative species (such as *Quercus*; Burns *et al.*, 1990). Current research suggests that leaf and wood phenology differ in their plasticity in response to changing environmental conditions across species and life stages (Dox *et al.*, 2022). Leaf phenology (e.g. leafout) in the spring is a highly plastic trait which varies substantially across years with different climates, tree age and also across species. Within a single community leafout can vary across species by over a month (Wolkovich *et al.*, 2014). How flexible growth is in response to changing growing season length and conditions varies between species and populations. Tree age may also impact responses, with younger trees being generally more flexible than their adult forms (Augspurger & Bartlett, 2003; Way & Montgomery, 2015) which could in part explain why they grew more with warmer seasons (Augspurger & Bartlett, 2003; Sweeney *et al.*, 2024). If our juvenile trees grew in a mature forest, they would have been positioned below the canopy, thus competing for light and other resources. Under these more constrained conditions, they might have not responded as much as we observed in our competition-limited conditions (Cleland & Wolkovich, 2024; Augspurger & Bartlett, 2003; Sweeney *et al.*, 2024; Polgar & Primack, 2011). In addition, we grew our trees in a common garden with potentially more space between trees than in a natural setting (spacing was between 1.5 and 3 m), without any shade cloth. While other studies on mature trees have found the same response, our results may have been more muted if the trees were grown in a forest setting.

Our focus on juvenile trees may suggest how forests will regenerate and assemble with the longer and warmer seasons of anthropogenic climate change. Plant phenology research suggested that earlier seasons with warming favor species that shift earlier the most, with many studies that focused on invasive species appearing to support this (Wolkovich *et al.*, 2013; Zettlemoyer *et al.*, 2019). Yet, these studies are often on herbaceous species, and our results, along with other recent work (Zani *et al.*, 2020; Zohner *et al.*, 2023), suggest mid- and later-season effects may underappreciate the performance implications in woody species. Thus, more studies of how later season events correlate with invasions, and forest assembly more generally, could help understand how warming will impact forest regeneration and range expansion.

Our results also suggest that population-level differences in assembly models may not be as critical as previously suggested (Cavender-Bares, 2019). While studies of large latitudinal gradients have found lower growth in trees sourced from poleward latitudes (Aitken & Bemmels, 2016; Soolanayakanahally *et al.*, 2013), we found little difference in growth across a latitudinal gradient of 3.5°. Thus, at least across this latitudinal range, local adaptation on growth appears weak. Supporting this, we found no evidence of a shifting end of season across provenances in these species, on average, despite decades of work finding that end of season events, such as budset, are usually more locally adapted than start of season events such as leafout (**?**Habjørg, 1978; **?**; Soolanayakanahally *et al.*, 2013; Aitken & Bemmels, 2016; Zeng & Wolkovich, 2024). Yet, a study of the same common garden including all tree and shrub species found that the end of season appeared to vary strongly by year and shift earlier with earlier start of season (Buonaiuto *et al.*, 2026), suggesting that the end of season may vary more than currently proposed (Alberto *et al.*, 2013). Taken together, these results imply that our current understanding of budset being strongly locally adapted, and thus varying little with temperature or the start of season, needs more study.

Implications for large-scale vegetation modeling

Leveraging a rarely available common garden dataset of tree growth coupled with leaf phenology for the start and end of season, we found an overall positive response from multiple tree species to warmer seasons, which has important implications for carbon models. While juvenile trees store less carbon than their adult forms, they are crucial for forest regeneration and range expansion (as discussed above, and see Luu *et al.*, 2024). In addition to their ecological role, juvenile trees generally predict adult tree responses qualitatively (Augspurger & Bartlett, 2003; Wolkovich *et al.*, 2025), despite certain ontogenic differences in stress residence and mortality rate across life stages (Caron *et al.*, 2021). Thus our results on young trees can likely inform how their adult counterparts will respond to changing growing season conditions.

We also found that the start and end of season asymmetrically matter to growth, which means that how we model these phenological events in vegetation and related earth system models may be critical for accurate carbon dynamic predictions (Li *et al.*, 2022). Such models usually assume a temperature-sensitive start of season and a photoperiod-controlled, and thus less variable, end of season (Chen, 2022; Melaas *et al.*, 2016). This results in a disproportionate influence of the start compared to the end of season on tree growth, unlike what our results suggest. Such a representation of what drives the temporal boundaries of the growing season may feed into oversimplified explanations of the processes regulating tree growth with future climate change.

In addition, we show important species differences (even across our closely related species) which are rarely accounted for in land surface models, despite this weakness being previously pointed out (Melaas *et al.*, 2016). These models still do not distinguish species and usually pool them together (Melaas *et al.*, 2016; Harrison *et al.*, 2021). This oversimplification could also mislead which species to target for reforestation, and forest types for conservation (Ruehr *et al.*, 2023; Chang *et al.*, 2024). Thus, better modelling approaches of the start and end of season will require more studies, on more species across a larger latitudinal gradient. Alongside improving land surface models, such studies could help tease out whether local adaptation of the end of season depends on the species, latitudinal range (both distance and how far north provenances come from), or some combination of the two.

Integrating our results into models will also require considering how environmental conditions beyond temperature, shift across the season, especially for soil moisture (Gao *et al.*, 2022; Li *et al.*, 2023). While trees may have a longer period to grow, water availability may become insufficient to support continuous growth throughout the growing season (Wang *et al.*, 2025). We regularly watered the trees in our common garden, likely relieving any major drought pressures, though studies beyond ours connected a later end of season and more growth (see: Gao *et al.*, 2022; Wang *et al.*, 2025). In the range of our four species, drought occurrence has not yet increased (National Centers for Environmental Information (NCEI), 2026; Agel *et al.*, 2025), suggesting that trees may grow more with longer and warmer growing seasons (Cook & Pederson, 2011; Agel *et al.*, 2025), as we found. The positive effect of a later end of season, however, could diminish if soil moisture becomes inadequate for growth, and other temperate regions with decreased soil moisture have reported declines in tree growth (e.g., Chiang *et al.*, 2021; Spinoni *et al.*, 2018). Improved accuracy of future carbon sequestration models would consequently require integrating over more regions, growth drivers and species.

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Supplemental material

Table S1: Number of individuals sampled per species and per provenance for a total of 81 individual trees sampled from the United States of America (in Massachusetts (MA) and New Hampshire (NH)) and Canada (in Quebec (Qc)).

Provenance	*A. incana* ($n = 498$)	*B. alleghaniensis* ($n = 29$)	*B. papyrifera* ($n = 20$)	*B. populifolia* ($n = 7$)
Dartmouth College (NH, USA)	116	8	7	2
Harvard Forest (MA, USA)	133	7	0	4
St-Hippolyte (Qc, Canada)	121	8	6	1
White Mountains (NH, USA)	128	6	7	0

Table S2: Provenance and common garden location (Arnold Arboretum) coordinates and 1951–1980 climate normals: mean annual temperature (MAT; °C), mean summer precipitation (MSP; mm), and frost-free period (FFP; days). We extracted these climate variables at each provenance from (Wang *et al.*, 2016). At the common garden location, the mean summer temperature was similar across years, with 2019 being the coolest (22.4°C), 2020 the hottest (23.1°C), and 2018 in between (22.9°C). There was no drought for those three years, except in 2020, when a moderate summer drought (June–August) ramped up to a severe drought in September (National Centers for Environmental Information (NCEI), 2026).

Provenance	Latitude	Longitude	Altitude	MAT	MSP	FFP
Arnold Arboretum (MA, USA)	42.3	-71.1	35	9.8	420	162
Harvard Forest (MA, USA)	42.5	-72.2	268	7.4	450	133
White Mountains (NH, USA)	44.5	-71.0	140	6.8	434	128
Dartmouth College (NH, USA)	44.8	-71.2	395	4.1	420	116
St-Hippolyte (Qc, Canada)	46.0	-74.0	371	3.4	472	121

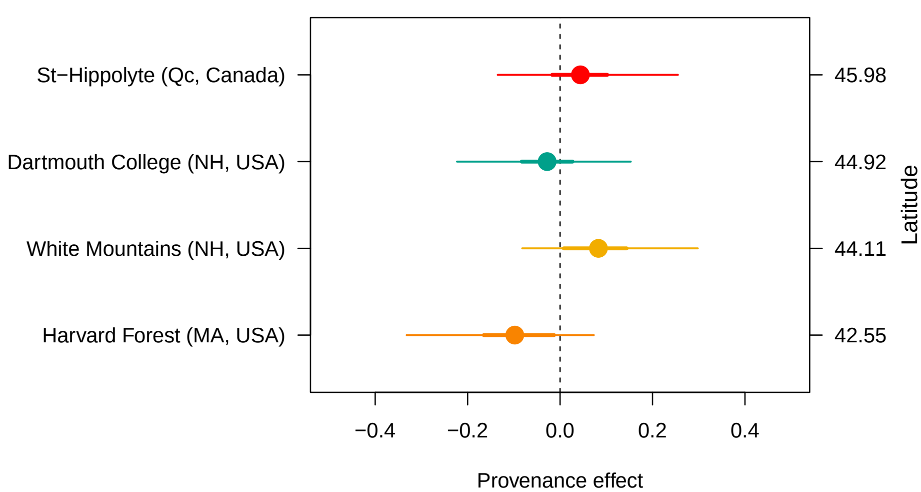

Figure S1: No effect of provenance on growth across a latitudinal gradient. We show the ring width (log(mm); see Table S3 for provenance parameter estimates) for each provenance with growing degree days as the predictor (effects were similar for the other predictors, see Tables S4, S5, S6). The dots represent the estimated mean of each provenance, with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively). The provenances are color-coded from Figure 2, the map showing their locations.

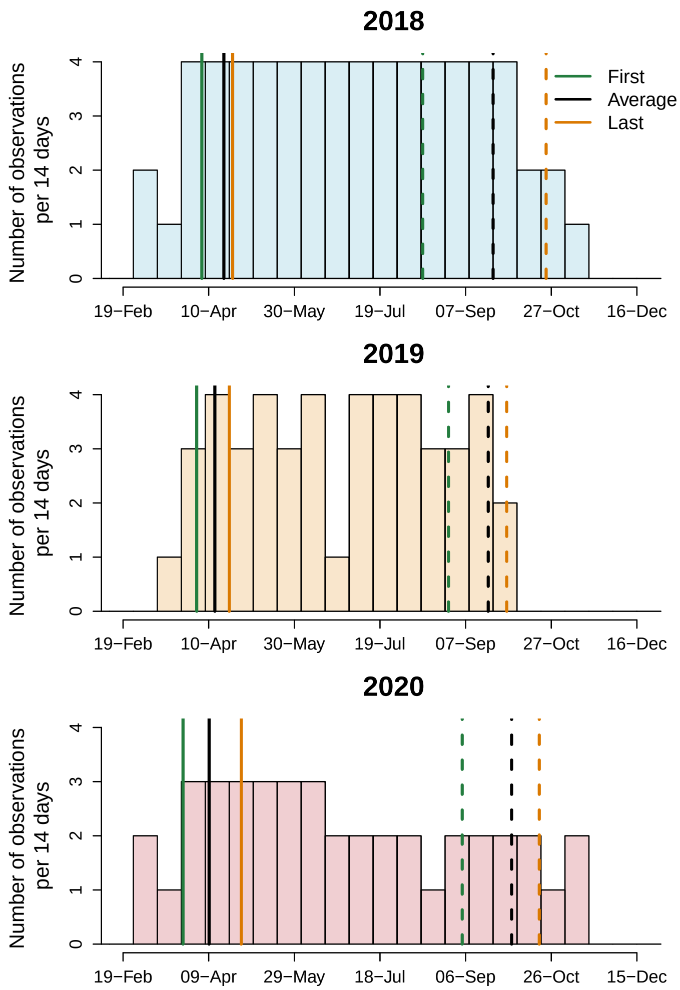

Figure S2: Leaf phenology monitoring frequency across years. We monitored leaf phenology once or twice per week for three years, though the frequency varied within and across years. Each histogram bin represents the number of observations that were made during a period of 14 days and the solid lines represent the earliest, the average and last recorded budburst observations; the dashed lines depict the budset observations. This demonstrates that across the different frequencies of observation, distances between early spring phenophases were similar across years.

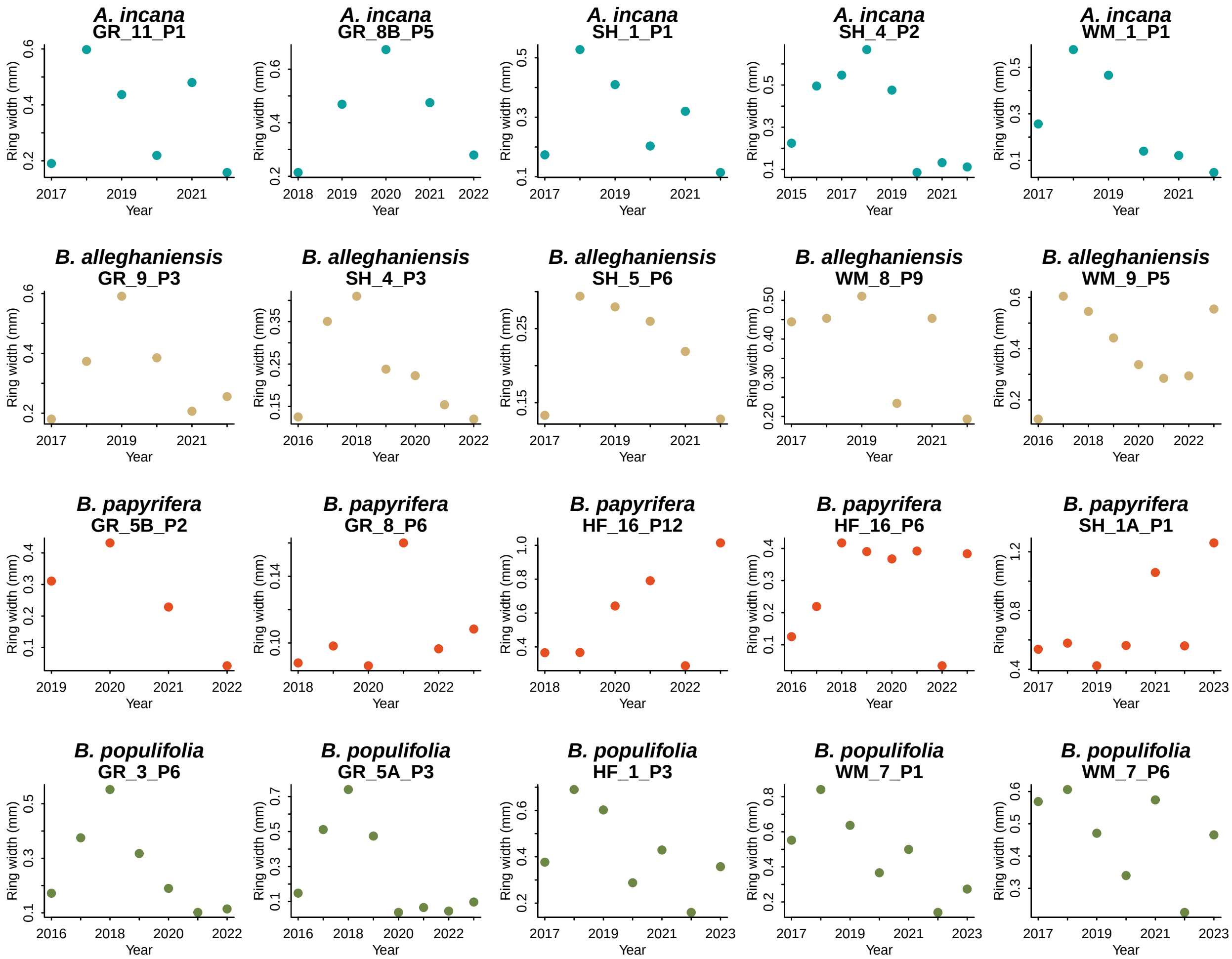

Figure S3: We show the potential allometry patterns across all years we had ring width for. For this, we randomly sampled 20 trees from our dataset and represent their ring width (mm) for each year.

Table S3: Summary output of each parameter for the growing degree days (GDD) predictor of ring width (log(mm)). We report estimates as mean and 90% and 50% uncertainty intervals. See 'Data analyses' in Methods section for more details.

Parameter	Mean	5%	25%	75%	95%
α	-0.57	-4.61	-2.30	1.14	3.57
$\sigma_{\alpha_{treeid}}$	0.17	0.05	0.13	0.21	0.27
$\sigma_{\alpha_{prov}}$	0.18	0.02	0.08	0.22	0.48
σ_y	0.48	0.43	0.46	0.50	0.53
$\beta_{species_{A.incana}}$	0.02	-0.01	0.01	0.04	0.05
$\beta_{species_{B.alleghaniensis}}$	0.04	0.01	0.03	0.06	0.08
$\beta_{species_{B.papyrifera}}$	0.15	0.06	0.11	0.18	0.23
$\beta_{species_{B.populifolia}}$	0.07	0.03	0.05	0.09	0.12
$\alpha_{prov_{DartmouthCollege(NH,USA)}}$	-0.03	-0.22	-0.08	0.03	0.15
$\alpha_{prov_{HarvardForest(MA,USA)}}$	-0.10	-0.33	-0.17	-0.01	0.07
$\alpha_{prov_{St-Hippolyte(Qc,Canada)}}$	0.04	-0.14	-0.02	0.10	0.26
$\alpha_{prov_{WhiteMountains(NH,USA)}}$	0.08	-0.08	0.01	0.14	0.30
$\alpha_{year_{2018}}$	0.17	-0.77	-0.22	0.56	1.14
$\alpha_{year_{2019}}$	0.16	-0.78	-0.24	0.55	1.12
$\alpha_{year_{2020}}$	-0.45	-1.39	-0.85	-0.06	0.51
$\alpha_{species_{A.incana}}$	1.01	-3.14	-0.69	2.74	5.18
$\alpha_{species_{B.alleghaniensis}}$	0.15	-4.04	-1.58	1.88	4.24
$\alpha_{species_{B.papyrifera}}$	-4.04	-9.02	-6.09	-2.03	0.89
$\alpha_{species_{B.populifolia}}$	-0.70	-5.04	-2.52	1.11	3.50

Table S4: Summary output of each parameter for the growing season length (GSL) predictor of ring width (log(mm)). We report estimates as mean and 90% and 50% uncertainty intervals. See 'Data analyses' in Methods section for more details.

Parameter	Mean	5%	25%	75%	95%
α	0.17	-3.84	-1.48	1.83	4.16
$\sigma_{\alpha_{treeid}}$	0.18	0.05	0.14	0.22	0.28
$\sigma_{\alpha_{prov}}$	0.18	0.02	0.07	0.22	0.48
σ_y	0.48	0.44	0.46	0.50	0.53
$\beta_{species_{A.incana}}$	0.05	0.00	0.03	0.07	0.10
$\beta_{species_{B.alleghaniensis}}$	0.03	-0.03	0.01	0.06	0.09
$\beta_{species_{B.papyrifera}}$	0.18	0.05	0.13	0.24	0.31
$\beta_{species_{B.populifolia}}$	0.06	-0.02	0.02	0.09	0.13
$\alpha_{prov_{DartmouthCollege(NH,USA)}}$	-0.04	-0.24	-0.09	0.02	0.15
$\alpha_{prov_{HarvardForest(MA,USA)}}$	-0.09	-0.32	-0.15	-0.01	0.08
$\alpha_{prov_{St-Hippolyte(Qc,Canada)}}$	0.04	-0.15	-0.02	0.10	0.25
$\alpha_{prov_{WhiteMountains(NH,USA)}}$	0.08	-0.08	0.01	0.14	0.30
$\alpha_{year_{2018}}$	0.22	-0.71	-0.16	0.61	1.16
$\alpha_{year_{2019}}$	0.09	-0.86	-0.29	0.48	1.03
$\alpha_{year_{2020}}$	-0.35	-1.29	-0.74	0.03	0.58
$\alpha_{species_{A.incana}}$	0.09	-3.97	-1.59	1.75	4.17
$\alpha_{species_{B.alleghaniensis}}$	0.33	-3.79	-1.34	1.99	4.40
$\alpha_{species_{B.papyrifera}}$	-2.71	-7.18	-4.56	-0.89	1.78
$\alpha_{species_{B.populifolia}}$	0.18	-3.92	-1.54	1.92	4.33

Table S5: Summary output of each parameter for the start of season (SOS) predictor of ring width (log(mm)). We report estimates as mean and 90% and 50% uncertainty intervals. See 'Data analyses' in Methods section for more details.

Parameter	Mean	5%	25%	75%	95%
α	1.08	-2.52	-0.44	2.56	4.77
$\sigma_{\alpha_{treeid}}$	0.21	0.11	0.18	0.25	0.30
$\sigma_{\alpha_{prov}}$	0.16	0.01	0.07	0.20	0.47
σ_y	0.47	0.43	0.45	0.49	0.52
$\beta_{species_{A.incana}}$	-0.07	-0.13	-0.10	-0.05	-0.01
$\beta_{species_{B.alleghaniensis}}$	0.10	0.00	0.06	0.14	0.20
$\beta_{species_{B.papyrifera}}$	0.05	-0.12	-0.02	0.12	0.21
$\beta_{species_{B.populifolia}}$	0.05	-0.07	0.00	0.10	0.16
$\alpha_{prov_{DartmouthCollege(NH,USA)}}$	-0.06	-0.27	-0.12	0.00	0.09
$\alpha_{prov_{HarvardForest(MA,USA)}}$	-0.04	-0.25	-0.10	0.01	0.12
$\alpha_{prov_{St-Hippolyte(Qc,Canada)}}$	0.02	-0.16	-0.03	0.08	0.21
$\alpha_{prov_{WhiteMountains(NH,USA)}}$	0.08	-0.07	0.01	0.14	0.30
$\alpha_{year_{2018}}$	0.21	-0.75	-0.17	0.59	1.18
$\alpha_{year_{2019}}$	0.12	-0.84	-0.26	0.49	1.08
$\alpha_{year_{2020}}$	-0.37	-1.34	-0.76	0.01	0.59
$\alpha_{species_{A.incana}}$	1.55	-2.16	0.10	3.03	5.10
$\alpha_{species_{B.alleghaniensis}}$	-1.85	-5.62	-3.39	-0.27	1.90
$\alpha_{species_{B.papyrifera}}$	-0.74	-5.04	-2.46	0.99	3.48
$\alpha_{species_{B.populifolia}}$	-0.47	-4.33	-2.02	1.12	3.31

Table S6: Summary output of each parameter for the end of season (EOS) predictor of ring width (log(mm)). We report estimates as mean and 90% and 50% uncertainty intervals. See 'Data analyses' in Methods section for more details.

Parameter	Mean	5%	25%	75%	95%
α	-1.40	-5.52	-3.10	0.32	2.70
$\sigma_{\alpha_{treeid}}$	0.19	0.07	0.15	0.23	0.28
$\sigma_{\alpha_{prov}}$	0.18	0.02	0.07	0.22	0.49
σ_y	0.47	0.42	0.45	0.49	0.52
$\beta_{species_{A.incana}}$	-0.01	-0.07	-0.04	0.01	0.04
$\beta_{species_{B.alleghaniensis}}$	0.12	0.05	0.09	0.14	0.18
$\beta_{species_{B.papyrifera}}$	0.21	0.09	0.16	0.26	0.33
$\beta_{species_{B.populifolia}}$	0.10	0.02	0.07	0.12	0.17
$\alpha_{prov_{DartmouthCollege(NH,USA)}}$	-0.04	-0.23	-0.09	0.02	0.14
$\alpha_{prov_{HarvardForest(MA,USA)}}$	-0.09	-0.33	-0.15	-0.01	0.07
$\alpha_{prov_{St-Hippolyte(Qc,Canada)}}$	0.04	-0.14	-0.02	0.10	0.25
$\alpha_{prov_{WhiteMountains(NH,USA)}}$	0.07	-0.09	0.01	0.14	0.28
$\alpha_{year_{2018}}$	0.23	-0.70	-0.16	0.62	1.17
$\alpha_{year_{2019}}$	0.11	-0.82	-0.28	0.51	1.07
$\alpha_{year_{2020}}$	-0.48	-1.41	-0.88	-0.09	0.47
$\alpha_{species_{A.incana}}$	3.18	-1.27	1.40	4.95	7.54
$\alpha_{species_{B.alleghaniensis}}$	-1.74	-6.24	-3.57	0.13	2.73
$\alpha_{species_{B.papyrifera}}$	-5.50	-11.08	-7.83	-3.20	0.08
$\alpha_{species_{B.populifolia}}$	-0.75	-5.40	-2.60	1.11	3.84

Table S7: Summary output of each parameter from our GDD (growing degree days) model, using basal area increment ($\log(mm^2)$) as the response variable. We report estimates as mean and 90% and 50% uncertainty intervals. See 'Data analyses' in Methods section for more details.

Parameter	Mean	5%	25%	75%	95%
α	0.62	-4.52	-1.50	2.77	5.84
$\sigma_{\alpha_{treeid}}$	0.52	0.41	0.47	0.56	0.64
$\sigma_{\alpha_{prov}}$	0.27	0.03	0.12	0.35	0.73
σ_y	0.63	0.57	0.61	0.66	0.70
$\beta_{species_{A.incana}}$	0.11	-0.03	0.05	0.17	0.25
$\beta_{species_{B.alleghaniensis}}$	0.20	0.02	0.13	0.26	0.36
$\beta_{species_{B.papyrifera}}$	0.92	0.49	0.74	1.10	1.34
$\beta_{species_{B.populifolia}}$	0.41	0.20	0.32	0.49	0.61
$\alpha_{site_{DartmouthCollege(NH,USA)}}$	-0.11	-0.47	-0.20	0.00	0.17
$\alpha_{site_{HarvardForest(MA,USA)}}$	-0.09	-0.45	-0.18	0.02	0.19
$\alpha_{site_{St-Hippolyte(Qc,Canada)}}$	0.06	-0.23	-0.04	0.15	0.42
$\alpha_{site_{WhiteMountains(NH,USA)}}$	0.15	-0.11	0.02	0.26	0.52
$\alpha_{year_{2018}}$	-0.36	-1.32	-0.77	0.03	0.61
$\alpha_{year_{2019}}$	0.33	-0.65	-0.07	0.73	1.32
$\alpha_{year_{2020}}$	-0.04	-1.01	-0.44	0.35	0.94
$\alpha_{species_{A.incana}}$	3.23	-2.25	0.98	5.48	8.55
$\alpha_{species_{B.alleghaniensis}}$	2.09	-3.40	-0.16	4.34	7.54
$\alpha_{species_{B.papyrifera}}$	-7.59	-14.84	-10.57	-4.73	-0.29
$\alpha_{species_{B.populifolia}}$	-0.07	-5.98	-2.41	2.29	5.57

Table S8: Summary output (β) of our GDD parameters (see Table S3 for untransformed parameter estimates), converted in % for a predicted extension in the start and end of season across, and for each species. We report estimates as posterior means with 90% and 50% uncertainty intervals.

Parameter	Mean	5%	25%	75%	95%
β_{SOS}	7.16	4.10	5.88	8.38	10.31
β_{EOS}	10.69	6.11	8.76	12.51	15.48
$\beta_{SOS_{A.incana}}$	1.72	-0.84	0.64	2.79	4.32
$\beta_{SOS_{B.alleghaniensis}}$	4.49	0.58	2.78	6.15	8.51
$\beta_{SOS_{B.papyrifera}}$	16.11	6.27	11.90	20.25	26.47
$\beta_{SOS_{B.populifolia}}$	6.31	2.19	4.54	7.98	10.63
$\beta_{EOS_{A.incana}}$	3.02	-1.46	1.12	4.90	7.63
$\beta_{EOS_{B.alleghaniensis}}$	7.18	0.92	4.41	9.85	13.71
$\beta_{EOS_{B.papyrifera}}$	22.58	8.61	16.50	28.46	37.58
$\beta_{EOS_{B.populifolia}}$	9.99	3.42	7.13	12.67	16.99

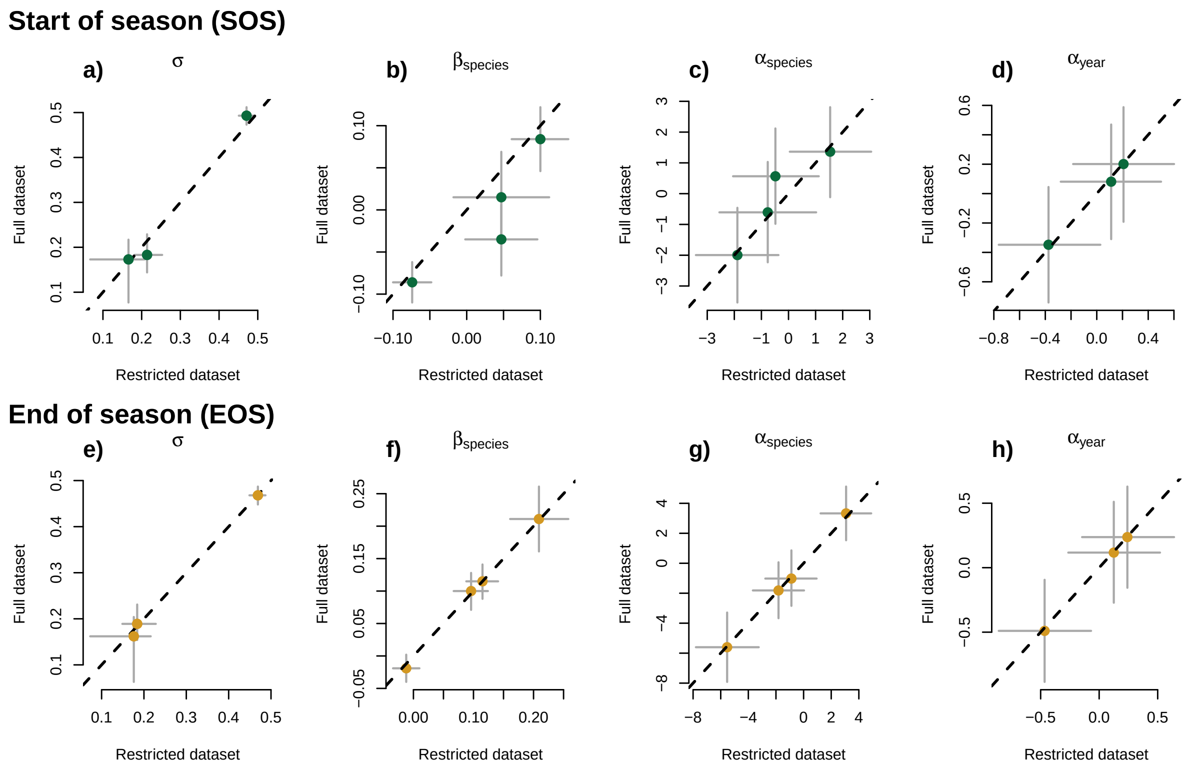

Figure S4: Full vs most restricted datasets for the model with start of season (a-d) and end of season (d-h) as the predictor. Each panel represent the different parameters used to fit the models. σ (a, e), $\alpha_{species}$ (c, g) and α_{year} (d,h) represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ (b, f) represent the change in ring width (log(mm))/adjusted predictor, for each species (see 'Data analyses' section in Methods for more details).

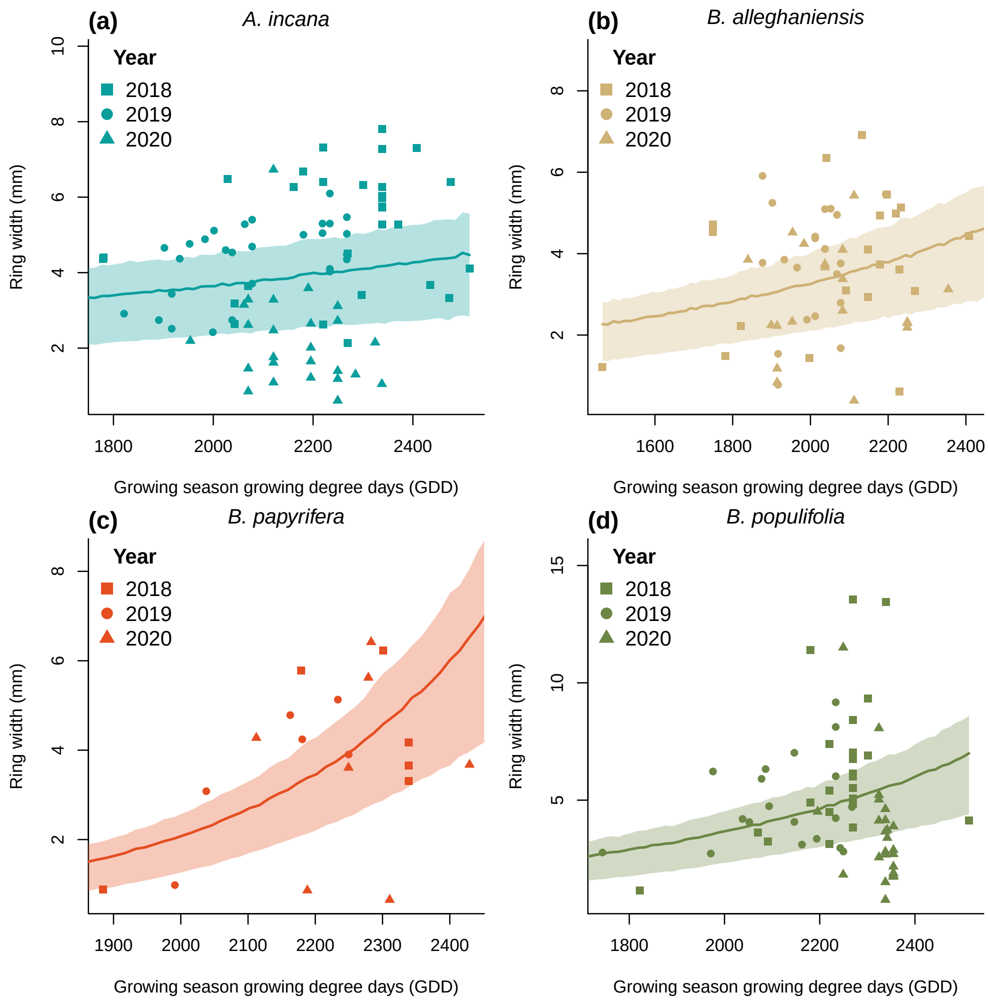

Figure S5: Ring width (mm) trends in response to thermal season (growing degree days). The lines represent the mean of the posterior distribution and the shaded area, the 50% uncertainty intervals at each predictor value back-converted from the model which uses ring width in log(mm) as the response variable. The dots are the empirical data shaped by year.

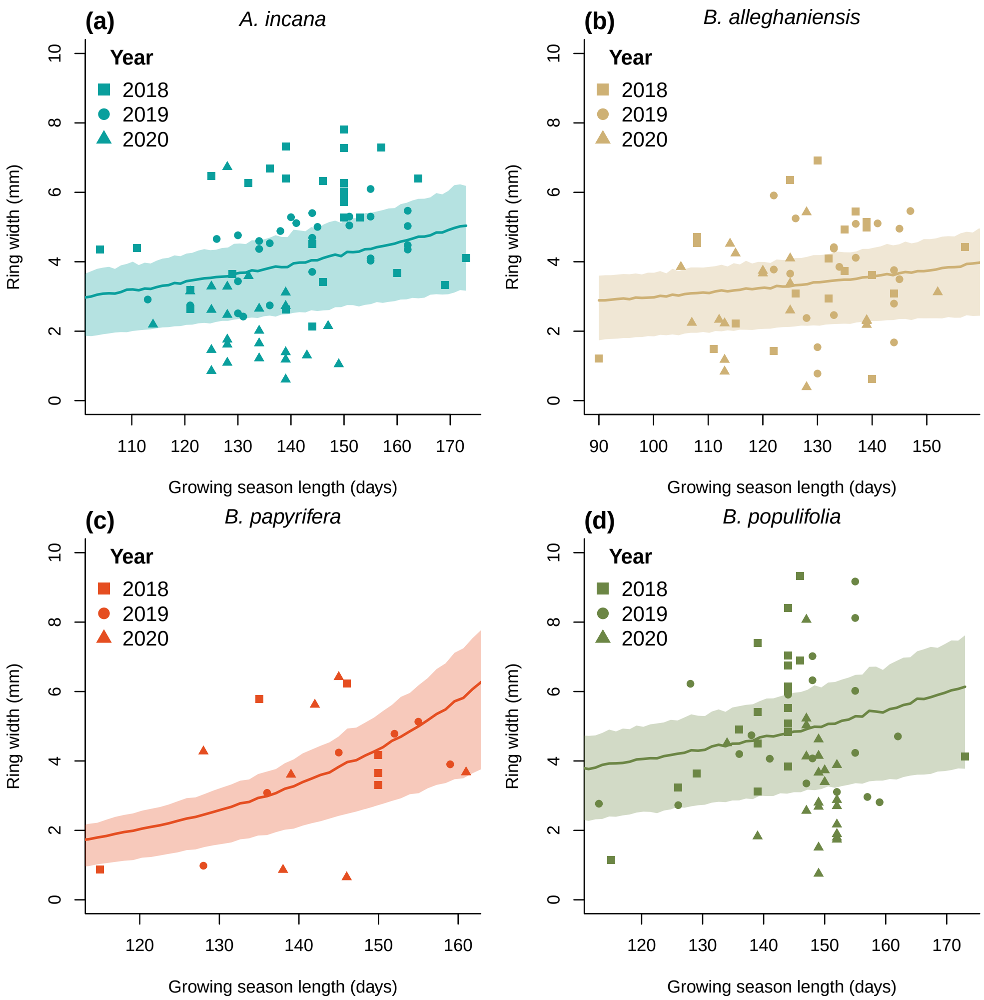

Figure S6: Ring width trends (mm) in response to calendar season (growing season length). The lines represent the mean of the posterior distribution and the shaded area, the 50% uncertainty intervals at each predictor value back-converted from the model which uses ring width in log(mm) as the response variable. The dots are the empirical data shaped by year.

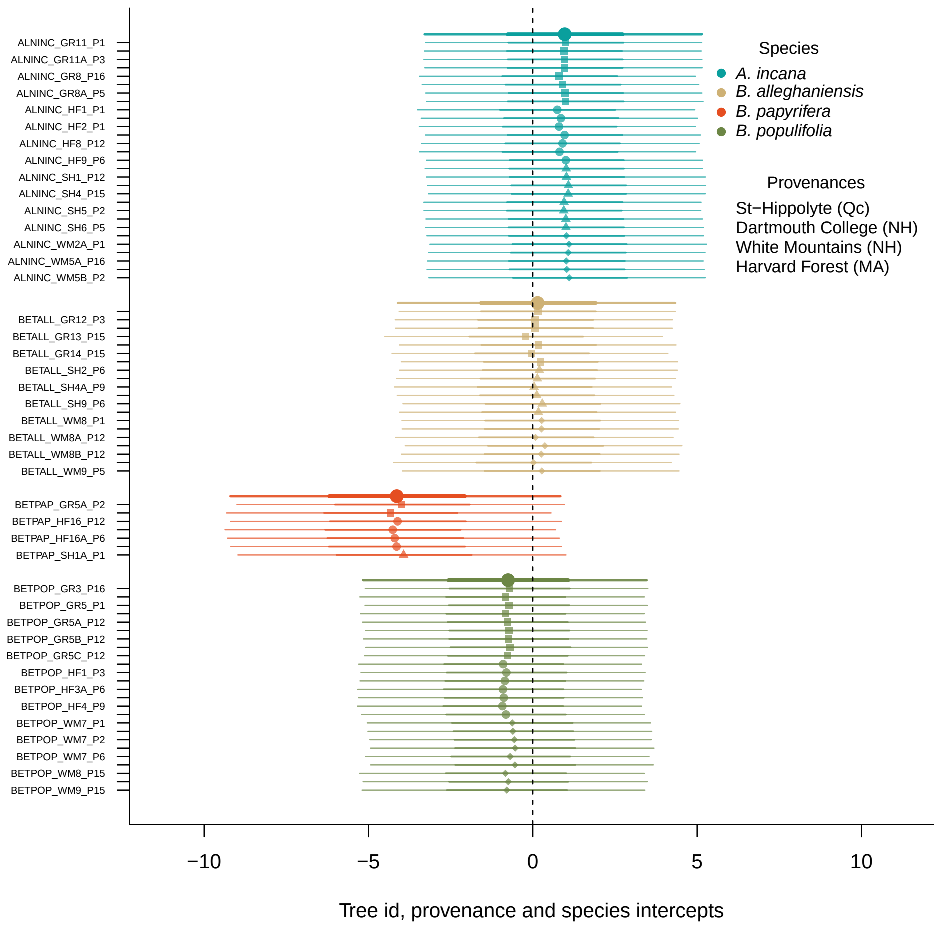

Figure S7: Ring width (log(mm)) deviations from the grand mean for grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 % and 90% uncertainty intervals.

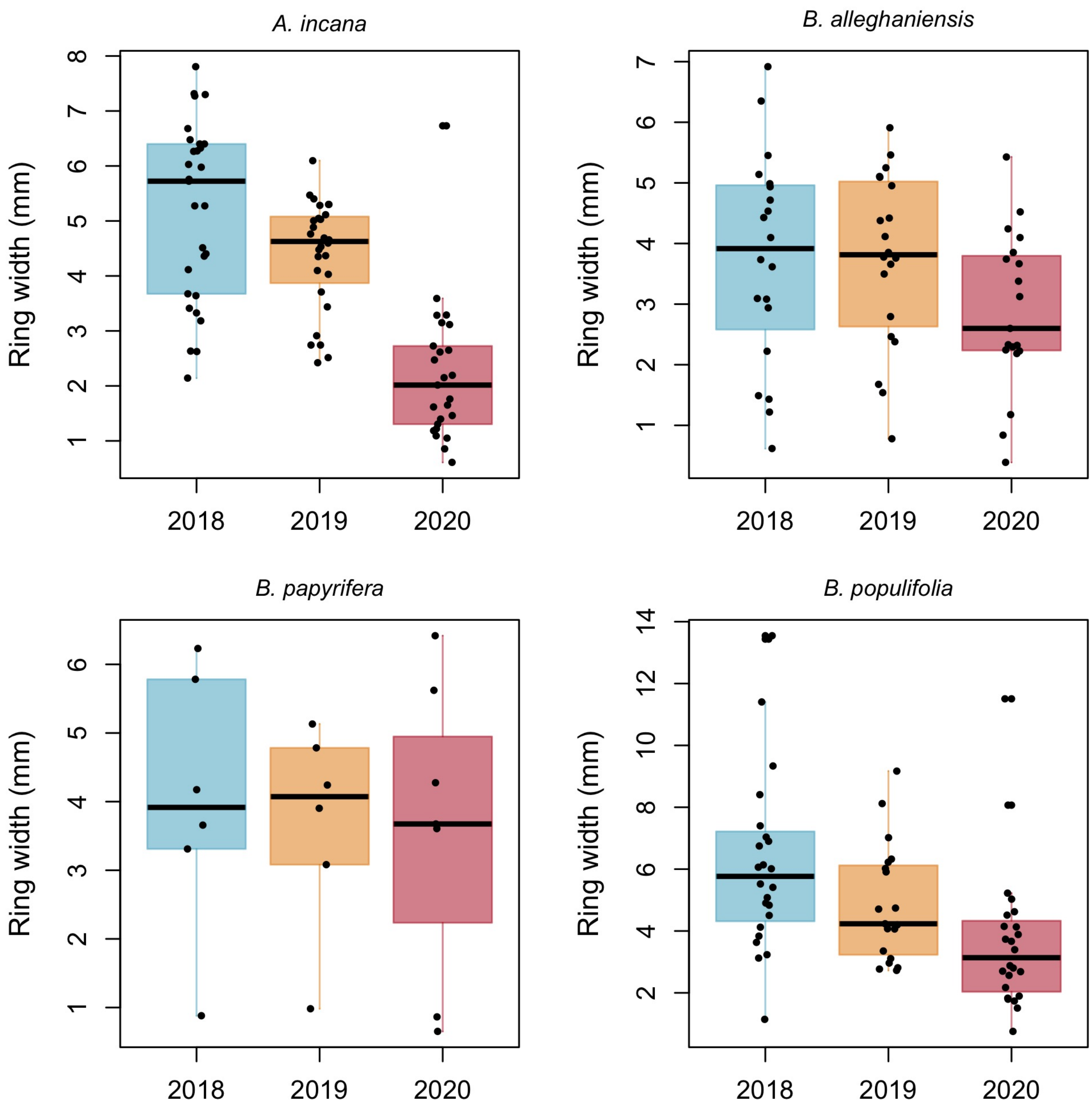

Figure S8: Boxplots representing the empirical data of our ring widths (mm) separated by species. The dots represent the raw data, the line is the mean and the box depicts the 50% uncertainty intervals.

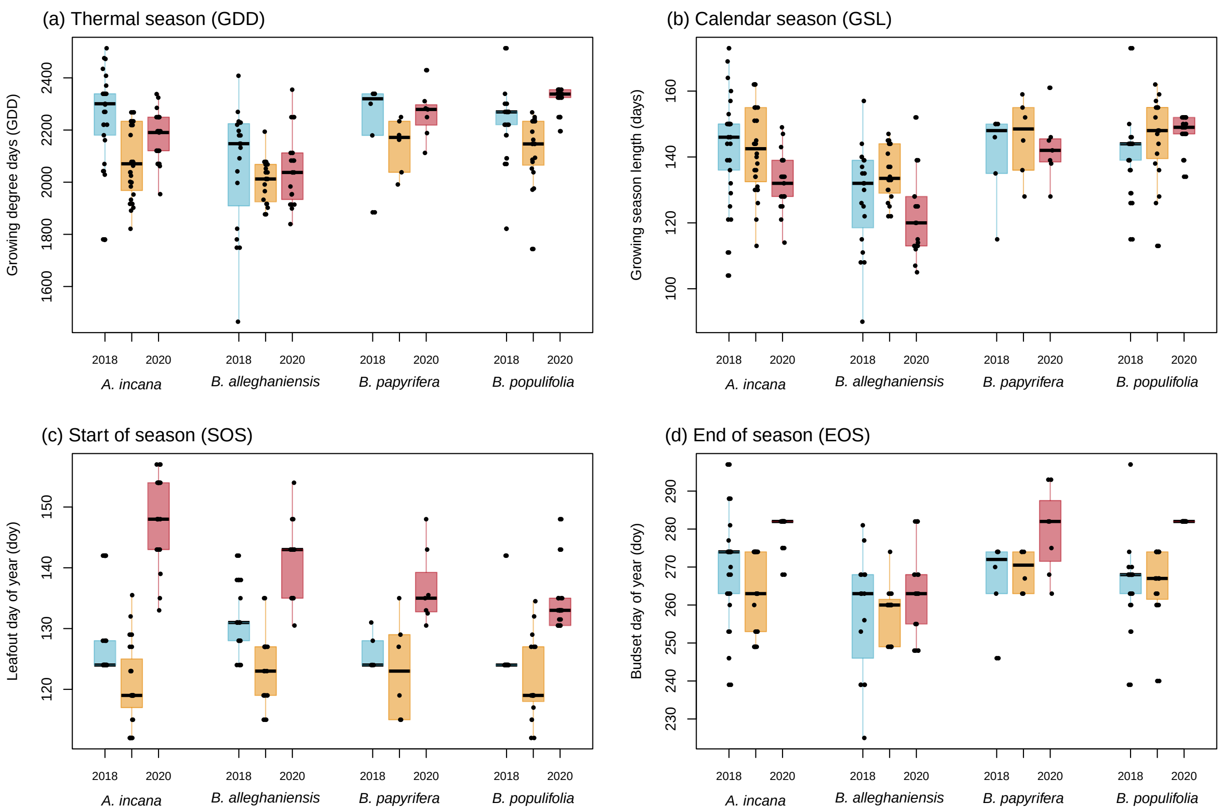

Figure S9: Boxplots representing the empirical data of all our predictors clustered by species, and for each year. The dots represent the raw data, the line is the mean and the box depicts the 50% uncertainty intervals.

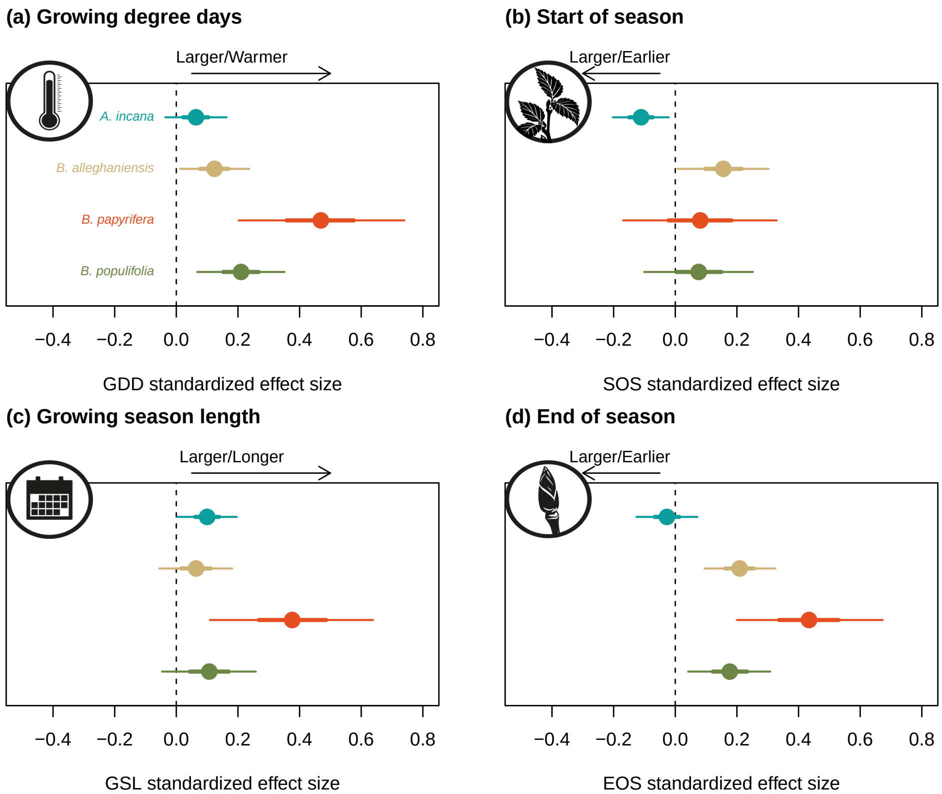

Figure S10: Effect size of each predictor across our four species. We depict ring width (log(mm)) change per adjusted predictor (z-scored; see 'Data analyses' section in Methods for more details) for each species. We report mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor (growing degree days, growing season length, start of season and end of season).)

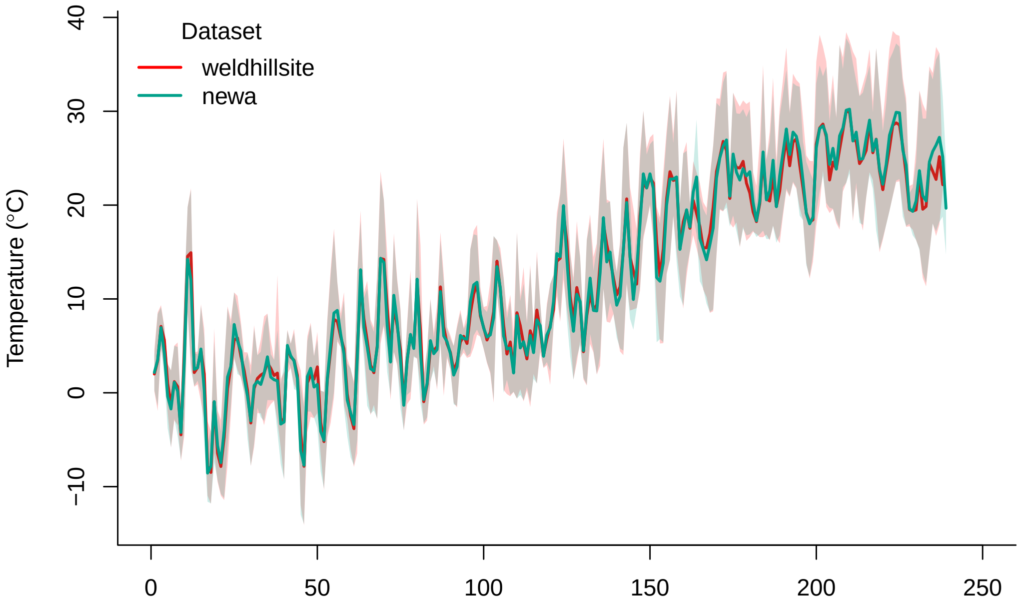

Figure S11: Climate data overlap in 2020 for the two climate stations we used in our dataset (see 'Data analyses' section for more details). The lines represent the daily mean temperature of each climate station and the shaded area, the minimum and maximum daily temperatures.